

Simulation of Soil Water within the Root Zone of a Corn Crop¹

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ABSTRACT

Knowledge of the distribution and movement of soil water in the root zone of a row crop is important in farm and watershed management, and in research on the transport of soil nutrients and pollutants. The objective of this study was to develop a model that would calculate the infiltration, evaporation, transpiration, and deep drainage of soil water in the root zone of a corn (*Zea mays* L.) crop throughout a growing season, using readily obtainable climatic data as daily inputs. The model differed from existing models in that the effect on the distribution of rainwater due to channeling in the cracks and non-capillary pores of a highly structured soil was considered. This distribution was obtained by adapting equations used to describe analogous distributions in ion exchange chromatography. Two methods of modeling plant uptake of soil water were compared. An empirical approach assumed that the plant took water from the layer in the root zone with the lowest energy level (determined four times daily) and the amount taken was given by a curve that related the ratio of actual and potential transpiration to the water content at that level. A mechanistic approach, similar to that used by Nimah and Hanks in 1973, considered root distribution, hydraulic conductivities, plant resistance, and root-soil contact resistance.

The model was tested using seasonal soil moisture records from 4 years on one soil and a single year on a second soil. Mean squared error of prediction for the seasons ranged from 0.05 to 0.15%. Use of the partial displacement infiltration component improved the accuracy of the results in all but one trial. The two methods of modeling plant uptake of soil water, mechanistic and empirical, differed very little in accuracy.

Additional index words: Partial displacement, modeling, Leaf area index.

THE ability to accurately calculate the distribution and movement of soil water within the root zone of a row crop is a prerequisite for effective irrigation management, modeling plant nutrient dynamics, estimating deep drainage recharge of ground water, and calculating nonpoint-source pollution rates of reservoirs. To this end, several models have been recently proposed (Childs et al., 1977; Davidson et al., 1978; Dutt et al., 1972; and Hillel, 1977); however, the effect of partial displacement due to channeling of rain water in non-homogeneous soils has not been considered. The objective of this study was to estimate the soil water distribution in the root zone of a corn (*Zea mays* L.) crop throughout the growing season by applying a model to data for two strongly structured soils.

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³Duncan, W. G., G. W. Thomas, and R. E. Phillips. 1974. Simulation of daily soil moisture profiles over four seasons using two tillage methods. Am. Soc. Agron. Abstr. p. 13.

STRUCTURE OF THE MODEL

A simplified flow diagram of the general model is given in Fig. 1. The four basic components are: potential evapotranspiration, leaf area index, infiltration and distribution of surface water, and uptake of water. Basic crop and soil parameters and daily meteorological data are read into the computer program, and the components (subroutines in the program) are activated during each one-day time step as follows: first, the leaf area index is computed using a procedure similar to that developed by Duncan et al.³; next, the evapotranspiration component, based on a model developed by Ritchie (1972), computes soil evaporation and plant evaporation (transpiration); the infiltration component adjusts the soil profile for surface water addition, using the model of Thomas et al. (1978), to describe the flow of surface water through well-structured soils; the uptake component then calculates the actual transpiration and adjusts the soil profile for surface evaporation and transpiration. Finally, the program computes and prints the water content at the end of each day for 15 cm increments down to 90 cm in the soil profile, and the cycle is repeated until the end of the growing season.

Evapotranspiration Component

The evapotranspiration component uses daily inputs of solar radiation in $\text{cal cm}^{-2} \text{ day}^{-1}$, maximum and minimum temperatures in C, and leaf area index to calculate the soil evaporation and potential plant transpiration for each day. It is adapted from Ritchie's model, which is based on an energy balance ap-

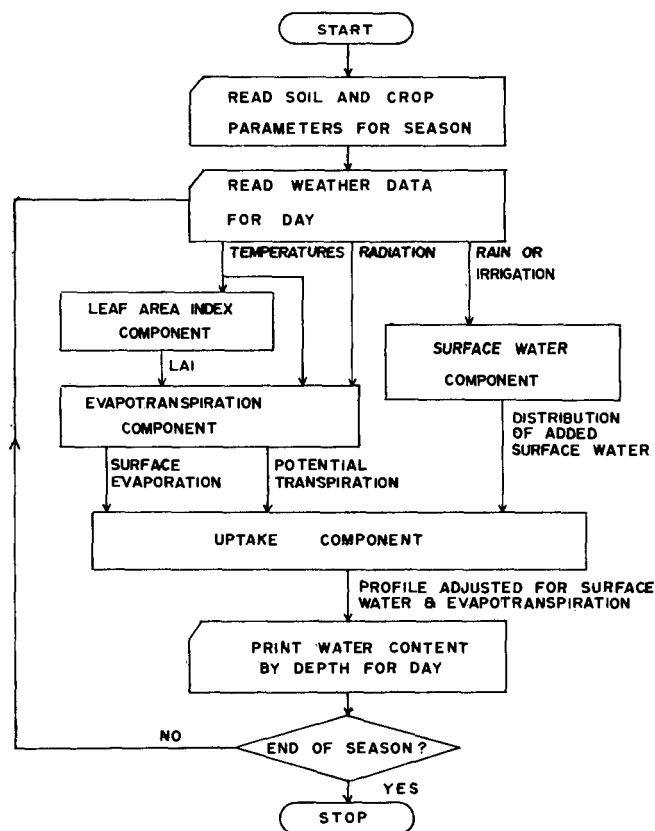


Fig. 1. Flow diagram of the model.

proach that assumes the amount of water evaporated from the surface and through the plant will be roughly equal to the potential energy available for evaporation as long as water is not limiting. The solar radiation reaching the canopy and soil surface is converted to latent heat of evaporation using simplified Penman (1948) equations. Evaporation from the soil is equal to the latent heat of evaporation as long as water is freely available at the surface. An amount of water characteristic of the soil is allowed to follow this freely evaporating rate. Below this limit, as a dry layer begins to form, evaporation is less than latent heat and is calculated using a falling rate, also characteristic of the soil. In a similar manner, water used in photosynthesis is ignored and transpiration is considered equal to total latent heat of evaporation less soil evaporation once full canopy (leaf area index ≥ 2.7) is achieved. For a developing canopy, an empirical equation relating latent heat, leaf area index, and transpiration is used. Soil evaporation and potential transpiration for the day, both in cm of water, are inputs to the uptake component.

Leaf Area Index Component

Rather than read in daily measured or tabular values of leaf area index (L_{ai}) as done in Ritchie's procedure, the proposed model uses a separate component based on the MAIZE model of Duncan et al.³ to compute the leaf area index of corn. Duncan's algorithm consists of an initial calculation of the maximum leaf area obtainable due to varietal and population considerations, and the use of growth curves to compute the daily change in leaf area as a function of physiological days (one physiological day = 21 degree days, with threshold temperatures of 10 and 30 C). The calculation of the maximum leaf area (L_{aim}) is accomplished using an equation that Duncan and Hesketh (1969) developed in phytotron studies with different races of corn. They found that the maximum leaf area index (L_{aim}) at plant populations below 29,640 plants ha⁻¹ was due to varietal considerations alone, but that at higher planting densities, the leaf area index increased with population. In the model, the maximum leaf area index due to varietal considerations (L_{ai+}) is computed by assuming leaf areas of 90, 110, and 130 dm² plant⁻¹ for early, intermediate, and full season maturity varieties, respectively. Leaf area is converted to leaf area index by multiplying by the quotient of the plant population and the number of dm²/ha. The maximum leaf area index, as a function of plant density (P), is then computed using the following equations:

$$L_{aim} = L_{ai+}; \text{ when } P < 29,640 \text{ plants ha}^{-1} \quad [1]$$

$$L_{aim} = 0.000668 L_{ai+} P^{0.71}; \text{ when } P \geq 29,640 \text{ plants ha}^{-1} \quad [2]$$

Duncan and Hesketh (1969) also found that the development of leaf area was closely related to accumulated physiological days, being exponential in the early stage, linear during the middle growth stage, reaching a maximum during tasseling and silking, and declining steadily thereafter. The slope of the linear portion of the curve was dependent on the maturity classification of the corn, being lower for varieties that matured later in the season. The three separate equations used in the MAIZE subroutine to describe the different types of leaf development were replaced with the following empirical equation for leaf growth up to and including tasseling:

$$L_a = L_{am} L_{ao} / [L_{ao} + (L_{am} - L_{ao}) \exp(-M t_p)] \quad [3]$$

where:

L_a = leaf area on a given day in dm² plant⁻¹

L_{am} = maximum leaf area in dm² plant⁻¹ = L_{aim} (10⁶ dm² ha⁻¹) (1/P)

L_{ao} = leaf area at seedling emergence = 0.2 dm² plant⁻¹

t_p = accumulated physiological days

M = maturity classification exponent (0.23, 0.20, and 0.17 physiological days⁻¹ for early, intermediate, and full maturity, respectively).

Ten physiological days after tasseling (which occurred at 55, 63, and 84 physiological days for early, intermediate, and full maturity in the model), leaf area was calculated by assuming a constant declining rate that was independent of physiological

days for the remainder of the growing season (Duncan et al.³).

$$L_a = L_{am} (1 - 0.005 t_c) \quad [4]$$

where: t_c = the number of calendar days after the decline started

The function of the leaf area index component was to make an initial calculation of the maximum leaf area using Eq. [1] or [2]. Thereafter, on a daily basis, the change in leaf area was computed using Eq. [3] or [4] depending on the stage of development. This leaf area was converted to leaf area index and used as an input to the evapotranspiration component.

Surface Water Component

It was assumed that the distribution of water in the soil profile, following the addition of surface water (rain in this case), could be described by the simple chromatographic equations of Thomas et al. (1978). They found that these equations accurately described the fraction of tagged water found at each depth following application of tagged artificial rainfall at the surface in a well structure soil. In the model, it was assumed that the initial distribution of surface water from rainfall or irrigation would be the same as that described for tagged water. The following equations were adapted from Thomas et al. (1978) to provide an empirical distribution of added water:

$$C = C_o; \quad X/X_t \leq K \quad [5a]$$

$$C = C_o [1/(1-K)] [(K X_t/X^{1/2} - K) X/X_t > K < X_t/X] \quad [5b]$$

$$C = 0 \quad X/X_t \geq 1/K \quad [5c]$$

where:

C = amount of water that is added at depth X in cm

C_o = field capacity at depth X in cm

X_t = (d/θ) expected depth of front with complete displacement

X = any depth of interest

θ = volumetric water content at field capacity

d = depth of water added in cm

K = coefficient of displacement ($0 \leq K \leq 1.00$)

These equations were used to calculate the addition of surface water at each level in the profile on days when precipitation occurred. In general, the amount of water added to each depth as a result of a rain or an irrigation event of d cm was calculated using Eq. [5a] or [5b] (depending upon the value of X/X_t). For soil depths greater than X_t/K , Eq. [5c] applied and no water was added. It was assumed that the water was added to each appropriate level and that displacement of the original soil moisture occurred only when the total water at a level exceeded field capacity for that level. A coefficient of displacement (K) of 0.05 was used in the model and this value agreed well with data previously obtained in the field. The component calculated the final profile distribution of water resulting from the rainfall or irrigation and passed this information to the uptake component.

Uptake Component

Due to the difficulty in measuring highly variable quantities, such as hydraulic conductivity and root surface area with depth, an empirical approach to uptake was investigated. The overall principle of this approach was the assumption that the plant optimized energy expenditures by taking water from the depth of lowest energy level (Wadleigh, 1945). The energy level at each depth on a given day was taken to be the hydraulic head potential (matrix potential plus the gravitational potential). The matrix potential was derived from experimentally determined potential-vs.-water-content relationships for each depth of a given soil. Since a single percent decrease in water content can change matrix potential by an order of magnitude, the relative order of energy levels may change as the transpirational water for a given day is withdrawn. To reduce this error, without incurring undue cost in computer time, the transpirational demand for each day was divided into four segments. A quarter of the total daily demand for water was taken from the depth with the least negative hydraulic head potential; then the water content at this depth was adjusted for the decrease, the relative

order of energy levels recalculated; the second quarter of water was withdrawn from the new optimum depth (which may or may not have been the same as for the first uptake), and so forth, through the remaining quarters. The total amount of water "demand" on a given day was calculated by the evapotranspiration component, based on climatic conditions. Actual transpiration, however, falls below potential as the water content in the profile decreases, slowing the rate of movement of water to the root and increasing the pathlength of flow. To account for this, a relationship was used that was similar to one proposed by Denmead and Shaw (1962) between water content and the fraction of potential water transpired:

$$E = E_o \cdot 0.0001 / [0.0001 + 0.9999 \exp(-1.45 \Theta_{aw})] \quad [6]$$

where:

E = actual transpiration in cm

E_o = potential transpiration in cm

Θ_{aw} = percent available water at the depth in question

For each day in the model, the potential transpiration (E_o , calculated by the evapotranspiration component) was divided into quarters; then the depth with the most favorable hydraulic head potential was selected, and the amount of actual water withdrawn from that level was a function of the available water in the layer as given by Eq. [6].

A mechanistic approach to plant uptake was also investigated to see if a higher degree of accuracy could be obtained. The procedure employed was based on an application of Darcy's law to plant transpiration wherein the total water transpired (Q in $\text{cm}^3 \text{ hour}^{-1}$) in a given time period at a given depth is proportional to the product of the difference in plant and soil potential ($H_p - H_s$ in cm), soil conductivity [$K(\theta)$ in cm hour^{-1}], and active root surface area [$\alpha R(t)$ in cm^2], divided by the distance of diffusion of the water to the root ($dr = 1.0$ cm), that is:

$$Q = [K(\theta)] [(H_p - H_s)/dr] \alpha R(t) \quad [7]$$

In the model the potential water transpired by the plant in a given hour was calculated by assuming that the potential transpiration for each day, given by the evapotranspiration subroutine, is sinusoidally distributed during the 12 daylight hours. The potential transpiration for an hour was equal to the sum of the fluxes at each depth i during the hour:

$$E_o = \sum Q_i = \sum [K_i(\theta)] [(H_p - H_{s,i})/dr_i] \alpha R_i(t) \quad [8]$$

It was assumed that the active root surface area $R_i(t)$ was equal to the estimated total root surface area at depth i , reduced by θ/θ_{sat} in accordance with the root contact model of Herkelrath et al. (1977). A root activity coefficient (α), empirically derived, of 0.10 was assumed so that Eq. [6] could be solved for plant potential (H_p) in the manner described by Nimah and Hanks (1973) and Hillel (1977). This plant potential is the lowest tension that will provide a sufficient gradient to extract enough water from the profile to satisfy the transpirational demand for that hour. The value of H_p was then used in Eq. [7] to solve for the amount of water, if any, extracted from each depth in an hour. It was assumed that the plant potential could not go below -15 bars. If a solution of Eq. [8] resulted in an H_p that exceeded this limit, the plant potential was set at -15 bars, the amount of water extracted from each depth was calculated, and in this case, actual transpiration was less than potential transpiration for the hour in question.

To complete the mechanistic approach, a root distribution function was obtained using a least-squares-fit of a plot of the root length data vs. time taken from Phillips et al. (1976). Unsaturated hydraulic conductivities at water contents above field capacity were measured with depth on the Maury silt loam using the method of Cassel (1975). An unsaturated conductivity function, $K(\theta)$, was calculated for each depth by assuming near-zero conductivity at wilting point and obtaining a linear least-squares-fit to a semi-log plot of the near-saturation and wilting point values vs. water content. Both the root distribution function and the conductivity function were obtained on the Maury silt loam.

In both approaches the uptake component adjusts each level as required for loss due to transpiration and the surface 15 cm for evaporative loss and passes the new moisture levels at the end of the day to the output section.

A listing of the FORTRAN IV computer program, as described, is available from the authors on request.

Sites and Measurements

The model was tested using corn ('Pioneer 3369A' — intermediate maturity class) on two different soils in various years. The moisture level in the field at 15 cm increments down to a depth of 90 cm was sampled gravimetrically approximately every 2 weeks during the growing season. Rainfall was measured on site using recording rain gauges. Maximum and minimum air temperatures were taken from the nearest weather station. Data on solar radiation for the experiments conducted in central Kentucky were taken from the weather station on the University of Kentucky Spindletop research farm, Lexington. With the exception of the Maury silt loam experiment in 1977, the initial soil profile water content was assumed to be at field capacity. In the case of the Maury in 1977, due to unusually low spring rainfall in the local area, it was assumed that the soil profile started at three quarters available water content.

The principal source of data was the long term conventional and no-tillage corn experiment at the Univ. of Kentucky research farm in Lexington. The soil is a Maury silt loam, 0 to 2% slope (Typic Paleudalfs, fine, mixed, mesic). Moisture samples were collected bimonthly from three 6.1×5.79 m conventional plots that received the same treatments (168 kg N $\text{ha}^{-1} \text{ year}^{-1}$ plus 11,250 kg lime in 1975) and averaged. Simulation was done for the years in which water content data had been collected: 1971, 1972, 1977, and 1978.

Data were also collected in 1977 from a corn crop grown in Shelby county, KY. The soil is a Shelbyville silt loam, 0 to 2% slope (Mollic hapludalfs, fine-silty, mixed, mesic). Moisture samples were taken bimonthly from two randomly-selected sites and averaged. The corn received 150 kg N in 1977. Since a no-tillage system was in use, soil evaporation in the model was multiplied by a factor of 0.8 to simulate the reduced evaporation due to the presence of a thin corn stover mulch.

The relative accuracy of the different models was measured by computing a mean squared error of prediction for the growing season of each simulation. This error consisted of the sum of the squared deviations between the simulated and the observed water contents divided by the total number of comparisons (number of sampling dates times number of layers).

RESULTS

The resultant mean squared errors using empirical and mechanistic approaches to uptake are shown in columns 1 and 2 of Table 1. Fig. 2 and 3 provide a comparison of the results obtained using the mechanistic and empirical approaches in 1971. Fig. 4 through 7 show the results for the remaining trials (when there was little difference in uptake scheme) using the empirical approach alone.

Seasonal mean squared errors were also computed for 3 years of simulation on the Maury silt loam using the soil water component (SOILWX) of Duncan's MAIZE model and are shown in column 5 of Table 1. This program assumes that infiltration of surface water is by piston flow rather than partial displacement and that water for plant transpiration is drawn from the layer in the profile nearest the soil surface containing available water. The evapotranspiration and leaf area index components are similar to those of the proposed model. Both empirical and mechanistic versions of the model using partial displacement fit the data better than SOILWX.

To ascertain the source of the increased accuracy, a component that used piston flow to distribute surface water was substituted in the model and the resulting mean squared errors are shown in columns 3 and 4 of Table 1. Comparison with columns 1 and 2 show that the results do not differ as much as expected and

Table 1. Comparison of mean squared error of prediction calculated from differences of actual and simulated soil water percentages of three soils for the empirical and mechanistic models when assuming partial displacement and piston flow of surface applied water.

Soil type and year	Partial displacement		Piston flow		SOILWX†
	Em- pirical	Mechan- istic	Em- pirical	Mechan- istic	
	mean squared error of prediction × 100				
Maury 1971	7.84	5.15	12.76	10.25	12.49
Maury 1972	14.61	15.37	9.31	10.69	26.45
Maury 1977	8.11	7.94	11.19	10.09	9.33
Maury 1978	8.08	9.15	12.41	13.35	
Shelbyville 1977	9.05		11.18		
Zanesville 1977	21.33		23.28		

† Calculated from soil water component (SOILWX) in the MAIZE model of Duncan et al. (1974).

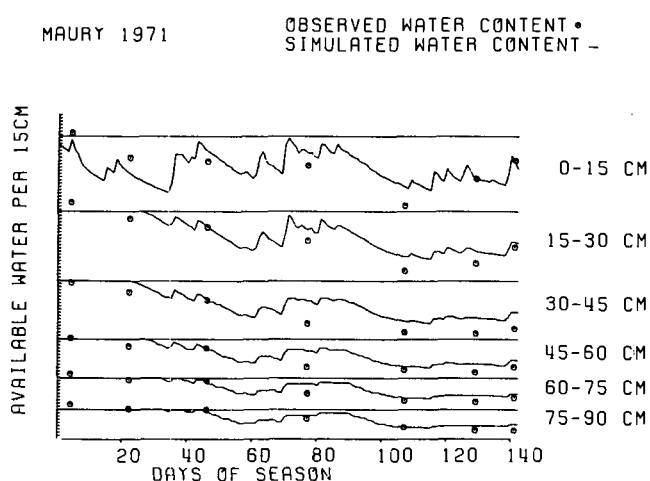


Fig. 2. Comparison of simulated and actual soil water contents in six layers of soil profile on the Maury silt loam in 1971 using the empirical uptake component. The two horizontal lines bracketing each 15 cm soil depth represent "field capacity" and wilting point, respectively. Plant available water, the difference in "field capacity" and wilting point, decreases with depth as indicated by the distance between horizontal lines. Each unit on the ordinate represents 1% water content on a volumetric basis.

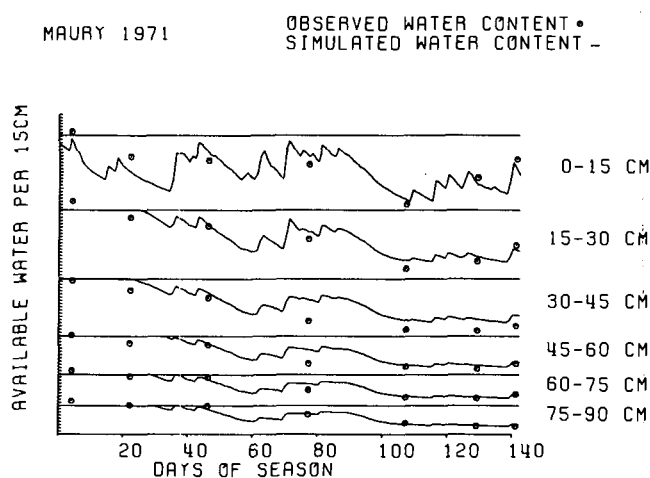


Fig. 3. Comparison of simulated and actual soil water contents on the Maury silt loam in 1971 using the mechanistic uptake component. See caption Fig. 2 for full explanation of graph.

Table 2. Comparison of number of daylight hours in which plant potential reached -15 bars out of total daylight hours in the season assuming partial displacement and piston flow of surface applied water.

Soil type and year	No. of daylight hours in which plant potential reached -15 bars		Total daylight hours
	Partial displacement	Piston flow	
Maury 1971	311	100	1,704
Maury 1972	457	363	1,212
Maury 1977	183	134	1,608
Maury 1978	292	224	1,452

MAURY 1972
OBSERVED WATER CONTENT •
SIMULATED WATER CONTENT -

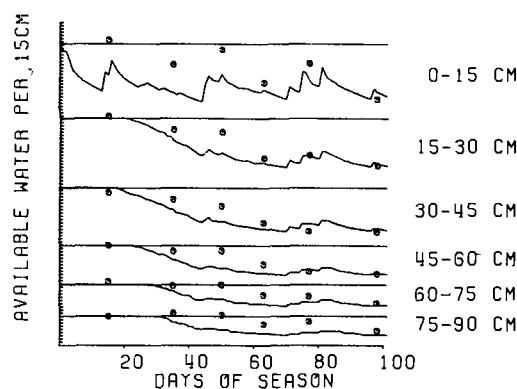


Fig. 4. Comparison of simulated and actual soil water contents on the Maury silt loam in 1972 using the empirical uptake component. See caption Fig. 2 for full explanation of graph.

that piston flow actually yields a better fit for 1972. It is apparent that a portion of the improved accuracy was due to the differences in the plant uptake components.

DISCUSSION AND CONCLUSIONS

The major difference between the proposed model and existing water models is the use of partial displacement in the addition of surface water. It was expected that this modification would be the principal source of any improvement in accuracy over other simulations. Although partial displacement does provide more accurate results than piston flow in all of the years, except 1972 (an especially dry year), the two schemes do not result in soil water profiles that are markedly different. The explanation may be that although partial displacement is important in describing the distribution of rain or irrigation water, it does not produce striking differences in the total profile water content when sampled at 2-week intervals. It is certain that piston flow puts more surface water into the top layers of soil, but this may not be discernible in bimonthly measurements, because this water is highly available and readily transpired. Similar profiles may result using piston flow (where the plant is freely transpiring the added water from the surface layers), and partial displacement (where water content in the top layers is lower, the plant is stressed more often, and total extraction of water from the pro-

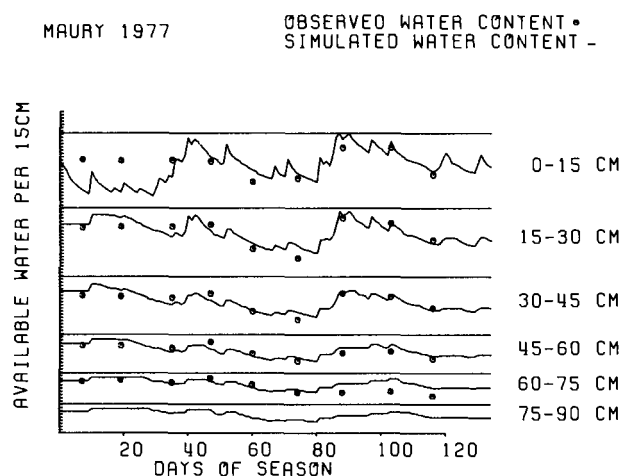


Fig. 5. Comparison of simulated and actual soil water contents on the Maury silt loam in 1977 using the empirical uptake component. See caption Fig. 2 for full explanation of graph.

file is reduced). Using the mechanistic approach to transpiration, a tally was made of the number of hours out of the total daylight hours in the season when the calculated -15 bar limit on plant potential was exceeded for partial displacement and for piston flow. Table 2 indicates that the plant was "stressed" considerably more often when assuming partial displacement than when assuming piston flow. In other words, although less surface water is added to the root zone with partial displacement, the total profile may not differ markedly from that obtained assuming piston flow, because plant stress has reduced transpiration.

The two different schemes for modeling plant uptake of transpirational water, empirical and mechanistic, also differ relatively little in accuracy, with the possible exception of 1971, in the initial trials with the Maury silt loam. However, lack of hydraulic conductivity and root distribution data on the Shelbyville soil precluded the use of the mechanistic model. This lack of flexibility coupled with higher simulation cost (\$7.10 as opposed to \$1.60 for 5 years data) make the mechanistic approach less attractive in applying the model to different conditions. On the other hand, the mechanistic approach has the potential for providing more insight into how the soil and plant interact. One example is the analysis of stress mentioned above and summarized in Table 2. In developing the model, it was also noted that a very dry surface layer and wet lower depths could result in a solution to Eq. [8] in which the plant potential was greater than the surface soil potential, reversing the gradient and "leaking" water into the surface layer from below during transpiration. This situation was avoided by setting conductivity equal to zero at the depth with the reversed gradient and resolving for plant potential.

The empirical approach to plant uptake also provided some insight into soil-plant interactions. For instance, the final shape of the curve relating the fraction of potential transpiration to water content in Eq. [6] was unexpected. Since the model was to be applied using field data, the curves that Ritchie (1973) obtained in experiments growing corn in a large weigh-

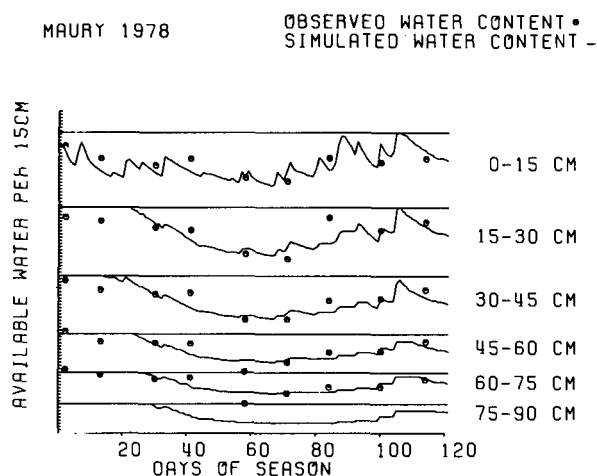


Fig. 6. Comparison of simulated and actual soil water contents on the Maury silt loam in 1978 using the empirical uptake component. See caption Fig. 2 for full explanation of graph.

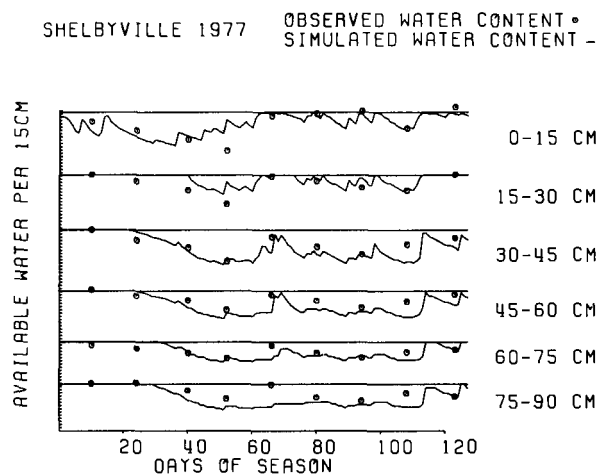


Fig. 7. Comparison of simulated and actual soil water contents on the Shelbyville silt loam in 1977 using the empirical uptake component. See caption Fig. 2 for full explanation of graph.

ing lysimeter were initially considered in preference to those produced by Denmead and Shaw (1962) from corn growing in pots. Ritchie's curves showed that the plant transpiration was insensitive to changes in water content except near the wilting point, whereas Denmead and Shaw's data showed much higher sensitivity to changes in water content throughout the range of available water. Ritchie has reasoned that the results differed because the corn roots grown in pots represented a static system that could not expand into unexploited wetter soil in response to decreasing water content. The roots in the lysimeter, on the other hand, were more dynamic and responded to a decreasing water content by proliferating into moist soil, consequently showing little response to changes in moisture for most of the range of available water. However, in this work, the response of the model was improved by using curves that are similar in shape to the highly sensitive curves that Denmead and Shaw found for corn grown in pots. The explanation may be that the shallow limestone-derived soils restrict root growth in the same way that pots do. In fact, data obtained by

Sookhakich (1976) indicates that corn root proliferation in the Maury silt loam is limited below 45 cm.

The assumption that the soil profile is at field capacity at the beginning of the growing season is normally valid for Kentucky. It is apparent that in 1977 on the Maury, however, assuming three-fourth available water initially was appropriate in all but the top layer, where the water content was even less (see Fig. 5). The initial profile water content that is assumed at the beginning of each season will depend on the general climatic characteristics of the locale and the specific conditions of the year being simulated. Ideally, an initial sampling of water content at planting would provide the best results.

Although the model was developed using a corn crop, it should also be applicable to other annual crops with similar leaf development such as sugarcane (*Saccharum officinarum* L.) and grain sorghum (*Sorghum bicolor* L.). Crops that have more horizontally configured leaves, such as soybeans, will require development of a component to generate leaf area index and some modification of the three empirical relationships in Ritchie's model that use leaf area index to calculate reflectance, the sunlight that filters through the canopy to the soil, and the potential transpiration at less than full canopy.

Application of the model to different types of soils will also require some modification. For more homogeneous, less structured soils, a higher K value used in Eq. [5] should be sufficient; sandy soils will require even larger K values, approaching piston flow. If the mechanistic approach to plant uptake is to be used, root distribution should be modified to show restriction in pan soils and expansion in deep uniform soils. If the empirical approach is used, a less sensitive "stress" curve, such as Ritchie's, would probably be appropriate in deep soils.

Although partial displacement was not as significant in modeling soil water content as expected, it has important implications in more complex models. For example, modeling movement, transformation, and plant uptake of soil N would require a distinction between newly-added rain water and residual soil water. Piston flow implies that the rain water is added to the surface and the residual water along with the soil N is displaced to a lower depth, when, in fact, in a well-structured soil much of the added water at deep depths is rain water that passed through channels and fissures in the surface without disturbing the N (Quisenberry and Phillips, 1976). Similar differences may be seen in modeling the rate of movement of pollutants into the ground water since with piston flow pollutants are slowly displaced downward through the soil mass with each rain, while under partial displacement a portion of the pollutant flows through channels to the water table in a single rain. The rate of simple ground water recharge and stream stage rise will also show significant differences. Recharge under piston flow

only occurs when the entire profile is at field capacity, so that water added at the surface by chain reaction displaces residual soil water into the underground streams. Using partial displacement, some amount of recharge occurs with almost every rain as a portion of the precipitation courses through the profile regardless of the water content.

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